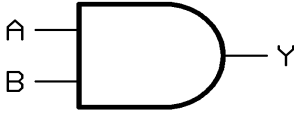
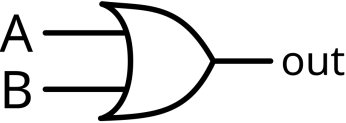
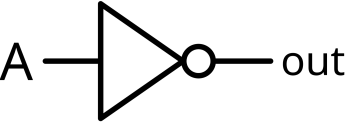
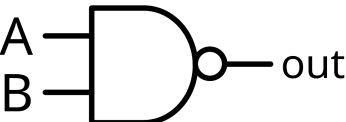
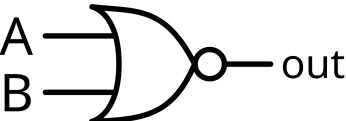
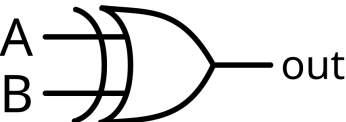
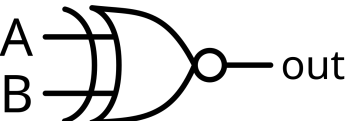


LAB-01
VERIFICATION OF LOGIC GATES

a) Aim: To verify logic gates

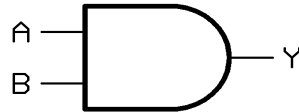
List of logic gates:

NAME	Symbol	IC NO
AND gate		7408
OR gate		7432
NOT gate		7404
NAND gate		7400
NOR gate		7402
XOR gate		7486
XNOR gate		74266

AND gate:

Description: The AND gate is a basic digital logic gate that implements logical conjunctions (A) from mathematical logic - AND gate behaves according to the truth table. A HIGH output (1) results only if all the inputs to the AND gate are HIGH(1).

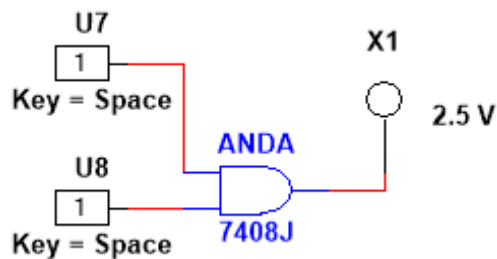
Circuit Diagram:



Truth table:

A	B	Y=A.B
0	0	0
0	1	0
1	0	0
1	1	1

Multisim circuit:



Components:

- 1)2 Interactive Digital Constants
- 2)AND Gate
- 3)LED
- 4)Ground

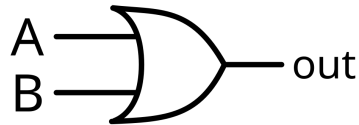
Procedure:

- 1.Open a multisim file.
- 2.create a new Circuit.
- 3.Place the components as in the circuit diagram.
- 4.Wire the circuit.
- 5.connect the LED to the ground.

OR gate:

Description: An OR gate is a fundamental digital logic gate that performs a logical OR operation .It has two or more input terminals and a single output terminal.The output of an OR gate is high(logic '1') if at least one of its input signals is high.If all input signals are low (logic '0'),then the output is low.

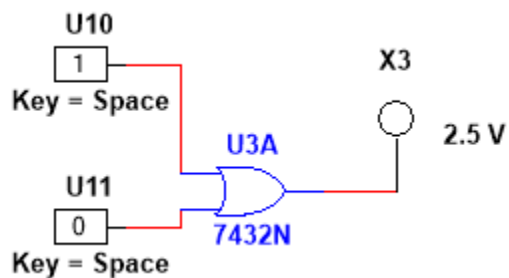
Circuit Diagram:



Truth table:

A	B	output=A+B
0	0	0
0	1	1
1	0	1
1	1	1

Multisim circuit:



Components:

2-Interactive Digital Constants,1-OR gate(7432),LED,Ground.

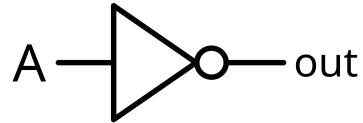
Procedure:

- 1.Open a multisim file.
- 2.create a new Circuit.
- 3.Place the components as in the circuit diagram.
- 4.Wire the circuit.
- 5.connect the LED to the ground.

NOT gate:

Description: A NOT gate is a logic gate that inverts the digital input signal.A NOT gate always has high (logical 1) output when its input is low (logical 0).Conversely, a logical NOT gate always has low(logical 0) output when the input is high(logical 1).

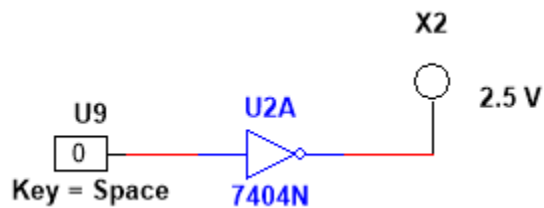
Circuit Diagram:



Truth table:

A	out
0	1
1	0

Multisim circuit:



Components:

2-Interactive Digital Constants, 1-NOT gate(7404), LED, Ground.

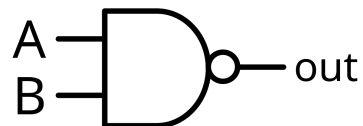
Procedure:

1. Open a multisim file.
2. create a new Circuit.
3. Place the components as in the circuit diagram.
4. Wire the circuit.
5. connect the LED to the ground.

NAND gate:

Description: A NAND gate ("not AND gate") is a logic gate that produces a low output(0) only if all its inputs are true, and high output(1) otherwise. Hence the NAND gate is the inverse of an AND gate, and its circuit is produced by connecting an AND gate to a NOT gate.

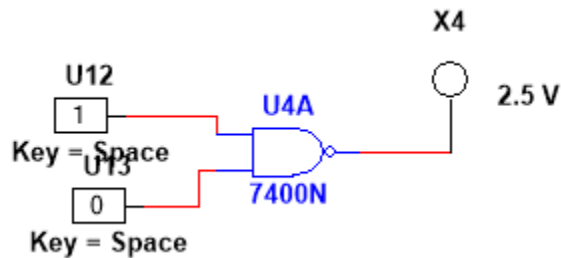
Circuit Diagram:



Truth table:

A	B	out
0	0	1

0	1	1
1	0	1
1	1	0

Multisim circuit:**Components:**

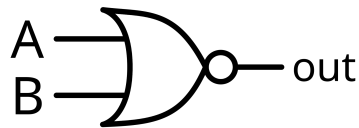
2-Interactive Digital Constants, 1-NAND gate(7400), LED, Ground.

Procedure:

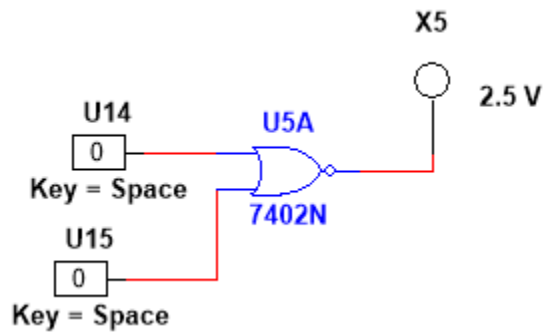
1. Open a multisim file.
2. create a new Circuit.
3. Place the components as in the circuit diagram.
4. Wire the circuit.
5. connect the LED to the ground.

NOR gate:

Description: A NOR gate ("not OR gate") is a logic gate that produces a high output(1) only if all its inputs are false, and low output(0) otherwise. Hence the NOR gate is the inverse of an OR gate, and its circuit is produced by connecting an OR gate to a NOT gate.

Circuit Diagram:**Truth table:**

A	B	out
0	0	1
0	1	0
1	0	0
1	1	0

Multisim circuit:**Components:**

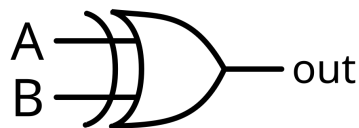
2-Interactive Digital Constants,1-NOR gate(7402),LED,Ground.

Procedure:

- 1.Open a multisim file.
- 2.create a new Circuit.
- 3.Place the components as in the circuit diagram.
- 4.Wire the circuit.
- 5.connect the LED to the ground.

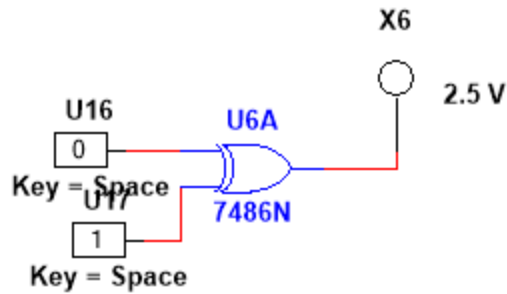
XOR gate:

Description: An XOR gate (also known as an EOR or EXOR gate) - pronounced as Exclusive OR gate - is a digital logic gate that gives a true(i.e a HIGH or 1)output when the number of true inputs is odd.

Circuit Diagram:**Truth table:**

A	B	out
0	0	0
0	1	1
1	0	1
1	1	0

Multisim circuit:

**Components:**

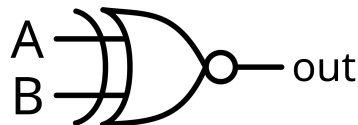
2-Interactive Digital Constants, 1-XOR gate(7486), LED, Ground.

Procedure:

1. Open a multisim file.
2. create a new Circuit.
3. Place the components as in the circuit diagram.
4. Wire the circuit.
5. connect the LED to the ground.

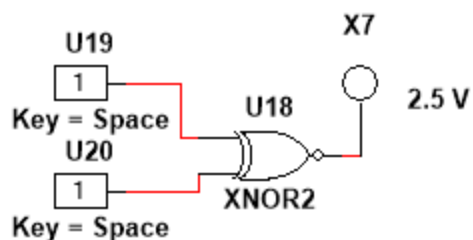
XNOR gate:

Description: The XNOR gate(also known as XORN'T,ENOR,EX NOR or N XOR)-and pronounced as Exclusive NOR - is a digital logic gate whose function is the logical complement of the exclusive OR gate(XOR gate).

Circuit Diagram:**Truth table:**

A	B	out
0	0	1
0	1	0
1	0	0
1	1	1

Multisim circuit:

**Components:**

2-Interactive Digital Constants, 1-NXOR gate(74266), LED, Ground.

Procedure:

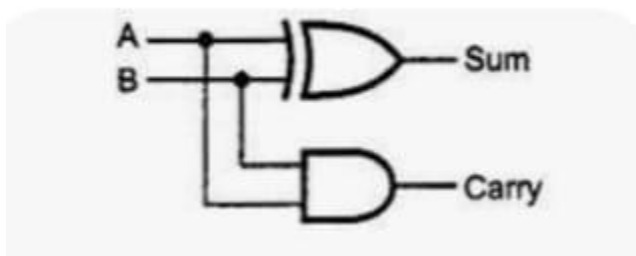
1. Open a multisim file.
2. create a new Circuit.
3. Place the components as in the circuit diagram.
4. Wire the circuit.
5. connect the LED to the ground.

LAB-02

a) Aim: To verify half adder, full adder, full adder using half adders, half subtractor, full subtractor, full subtractor using half subtractors

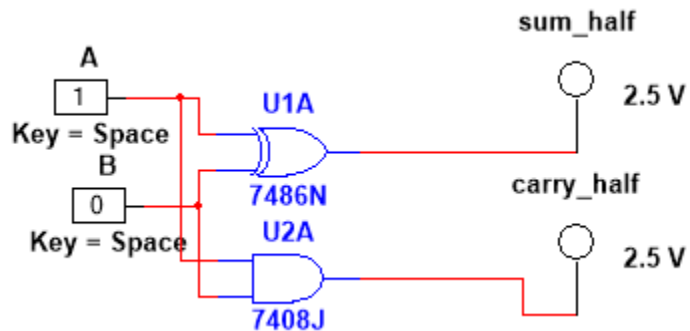
HALF ADDER:

Description: A half adder is a type of adder, an electronic circuit that performs the addition of numbers. The half adder is able to add two single binary digits and provide the output plus a carry value.

Circuit diagram:**Truth table:**

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0

1	1	0	1
---	---	---	---

Multisim circuit:**Components required:**

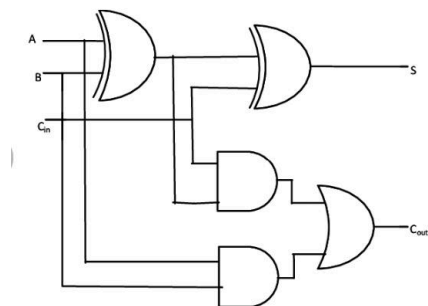
Two interactive digital constants, XOR gate, AND gate, two LEDs, Two grounds

Procedure:

- 1) Open multisim software
- 2) Create new design
- 3) Place the components and connect them according to the circuit diagram
- 4) Verify the circuit diagram with the help of truth table

FULL ADDER:

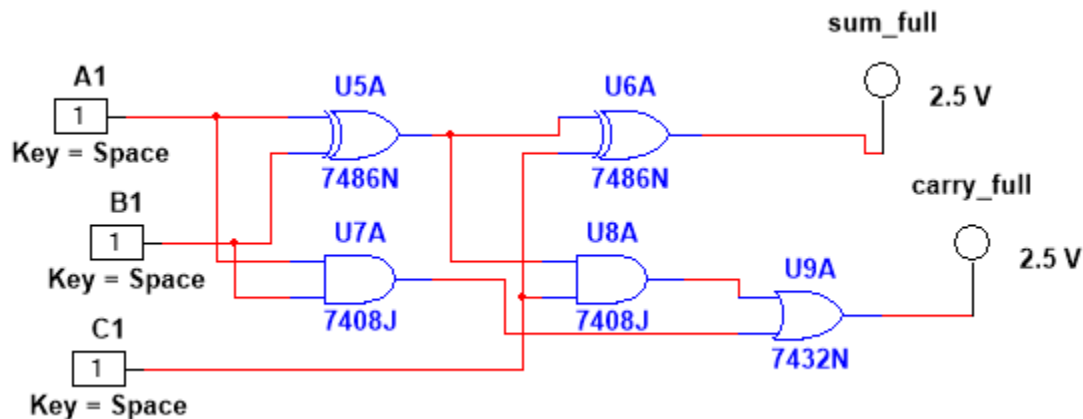
Description: A full adder adds three one-bit binary numbers, two operands and a carry bit. The adder outputs two numbers, a sum and a carry bit.

Circuit Diagram:**Truth table:**

A	B	C	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0

0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Multisim circuit:



Components required:

Three interactive digital constants, 2-XOR gate, 2-AND gate, 1-OR gate, two LEDs, Two grounds

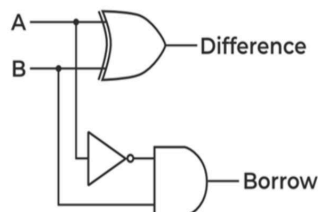
Procedure:

- 1) Open multisim software
- 2) Create new design
- 3) Place the components and connect them according to the circuit diagram
- 4) Verify the circuit diagram with the help of truth table

HALF SUBTRACTOR:

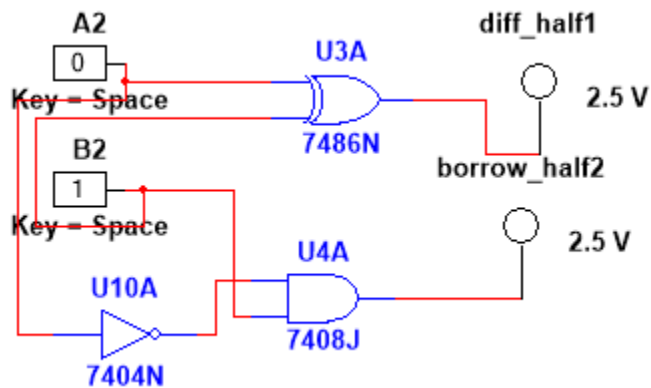
Description: The half subtractor is a combinational circuit which is used to perform subtraction of two bits. It has two inputs, A (minuend) and B (subtrahend) and two outputs Difference and borrow.

Circuit Diagram:



Truth table:

A	B	DIFF	BORROW
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Multisim circuit:**Components required:**

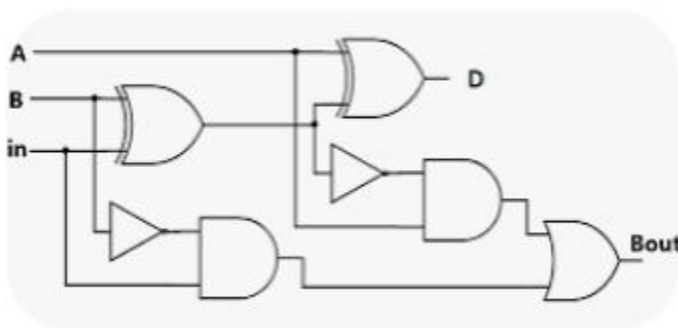
Two interactive digital constants, XOR gate, AND gate, NOT gate, two LEDs, Two grounds

Procedure:

- 1) Open multisim software
- 2) Create new design
- 3) Place the components and connect them according to the circuit diagram
- 4) Verify the circuit diagram with the help of truth table

FULL SUBTRACTOR:

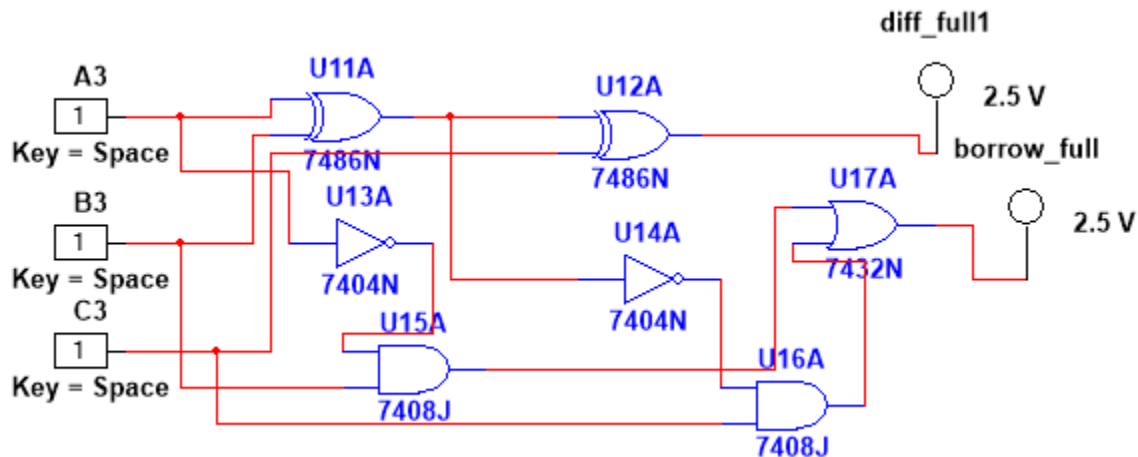
Description: The full subtractor is used to subtract three 1-bit numbers A, B and C which are minuend, subtrahend, and borrow, respectively.

Circuit diagram:

Truth Table:

A	B	C	DIFF	BORROW
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Multisim circuit:

**Components required:**

Three interactive digital constants, 2-XOR gate, 2-AND gate, 1-OR gate, two LEDs, Two grounds, 2-NOT gates.

Procedure:

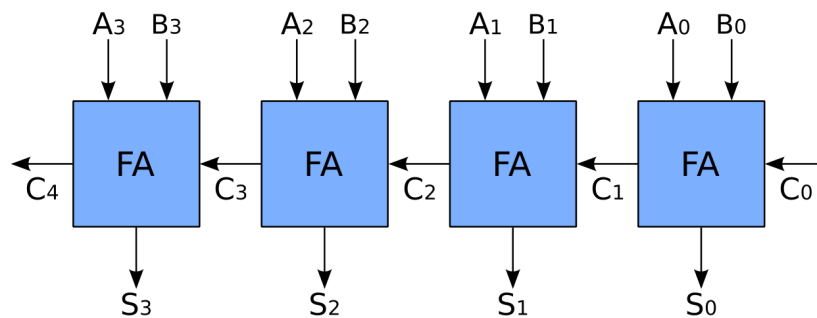
- 1) Open multisim software
- 2) Create new design
- 3) Place the components and connect them according to the circuit diagram
- 4) Verify the circuit diagram with the help of truth table

LAB-03

a)**Aim:** To verify ripple carry adder and carry look ahead adder.

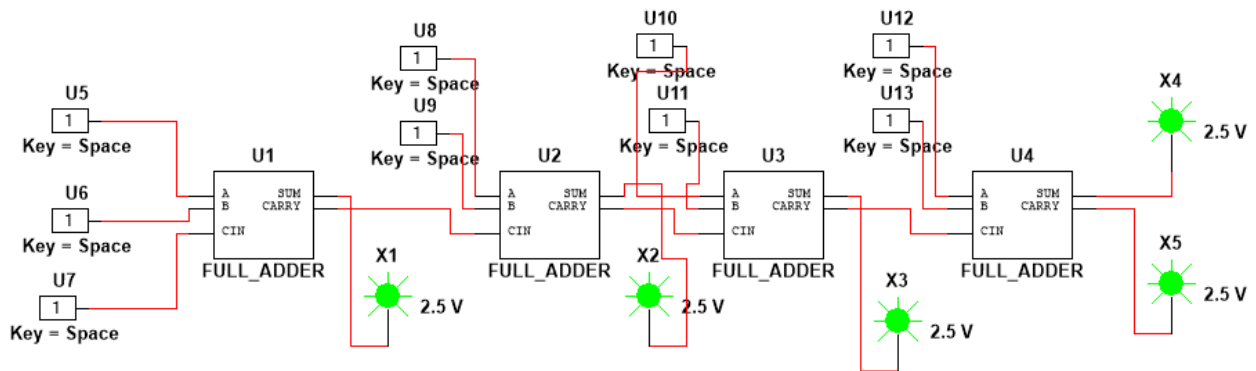
RIPPLE CARRY ADDER:**Description:-**

Ripple carry adder is a combinational logic circuit. It is used for the purpose of adding two n-bit binary numbers. It requires n full adders in its circuit for adding two n-bit binary numbers. It is also known as n-bit parallel adder.

Circuit diagram:-**Truth table:-**

cin	A3	A2	A1	A0	B3	B2	B1	B0	S3	S2	S1	S0	C out
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	1	0	0	1	0	0
0	0	0	1	0	0	0	1	0	0	1	0	0	0
0	0	0	1	1	0	0	1	1	0	1	1	0	0
0	0	1	0	0	0	1	0	0	1	0	0	0	0
0	0	1	0	1	0	1	0	1	1	0	1	0	0
0	0	1	1	0	0	1	1	0	1	1	0	0	0
0	0	1	1	1	0	1	1	1	1	1	1	0	0
0	1	0	0	0	1	0	0	0	0	0	0	0	1
0	1	0	0	1	1	0	0	1	0	0	1	0	1
0	1	0	1	0	1	0	1	0	0	1	0	0	1

0	1	0	1	1	1	0	1	1	0	1	1	0	1
0	1	1	0	0	1	1	0	0	1	0	0	0	1
0	1	1	0	1	1	1	0	1	1	0	1	0	1
0	1	1	1	0	1	1	1	0	1	1	0	0	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1

Multisim circuit:-**Components required:-**

Four full adders, five probes, interactive digital constants

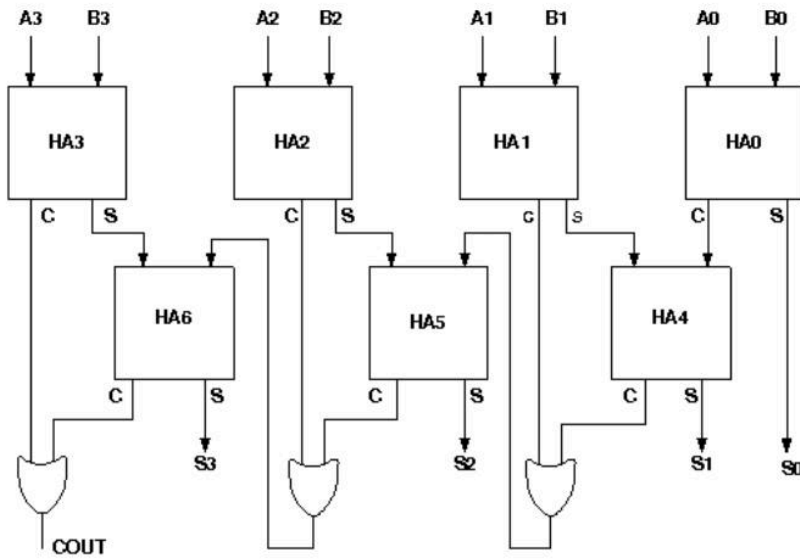
Procedure:-

- 1) open multisim software
- 2) create new design
- 3) place the components and connect them according to the circuit diagram
- 4) Verify the circuit diagram with the help of the truth table.

CARRY LOOK AHEAD ADDER:-**Description:-**

A carry-look ahead adder(CLA) is another type of carry propagate adder that solves this problem by dividing the adder into blocks and providing circuitry to quickly determine the carry out of a block as soon as the carry is known.

Circuit diagram:-

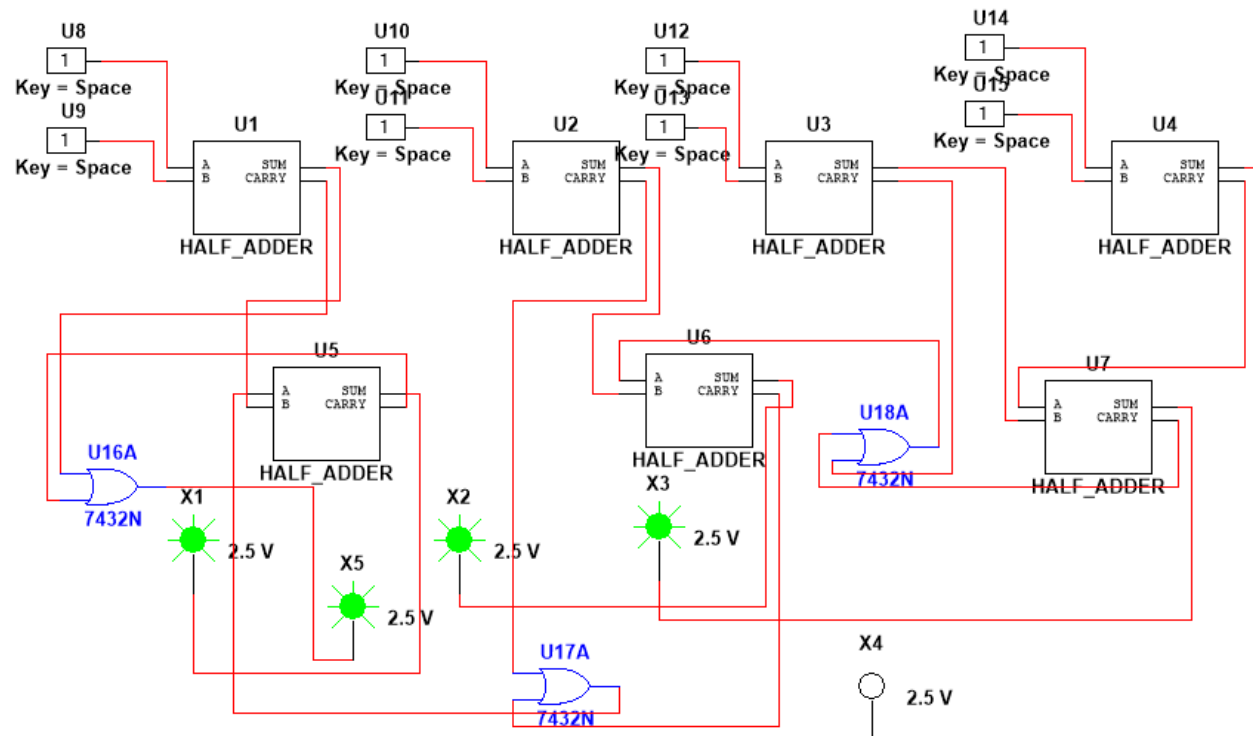


Truth table:-

cin	A3	A2	A1	A0	B3	B2	B1	B0	S3	S2	S1	S0	C out
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	1	0	0	1	0	0
0	0	0	1	0	0	0	1	0	0	1	0	0	0
0	0	0	1	1	0	0	1	1	0	1	1	0	0
0	0	1	0	0	0	1	0	0	1	0	0	0	0
0	0	1	0	1	0	1	0	1	1	0	1	0	0
0	0	1	1	0	0	1	1	0	1	1	0	0	0
0	0	1	1	1	0	1	1	1	1	1	1	0	0
0	1	0	0	0	1	0	0	0	0	0	0	0	1
0	1	0	0	1	1	0	0	1	0	0	1	0	1
0	1	0	1	0	1	0	1	0	0	1	0	0	1
0	1	0	1	1	1	0	1	1	0	1	1	0	1
0	1	1	0	0	1	1	0	0	1	0	0	0	1

0	1	1	0	1	1	1	0	1	1	0	1	0	1
0	1	1	1	0	1	1	1	0	1	1	0	0	1
0	1	1	1	1	1	1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1

Multisim circuit:-



Components required:-

Interactive digital constants, 7 half adders, 5 probes, 3 OR gates.

Procedure:-

- 1) open multisim software
- 2) create new design
- 3) place the components and connect them according to the circuit diagram
- 4) Verify the circuit diagram with the help of the truth table.

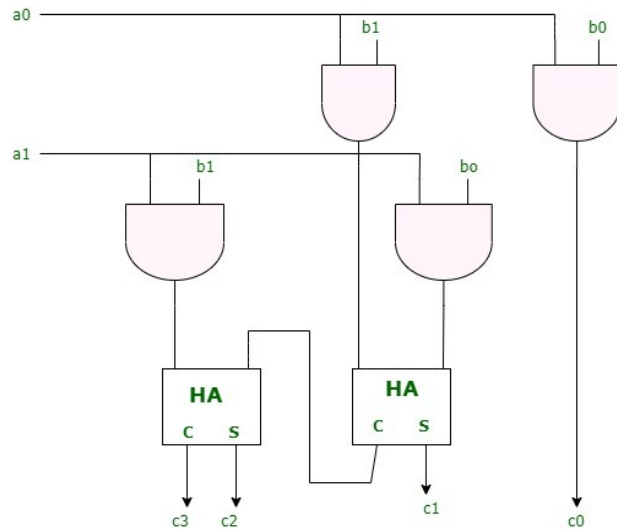
LAB-04

a)Aim: To verify 2 by 2 array multiplier

2 by 2 array multiplier:

Description: A 2 by 2 array multiplier is a digital circuit used in computer arithmetic to perform multiplication of two binary numbers, each represented by two bits.

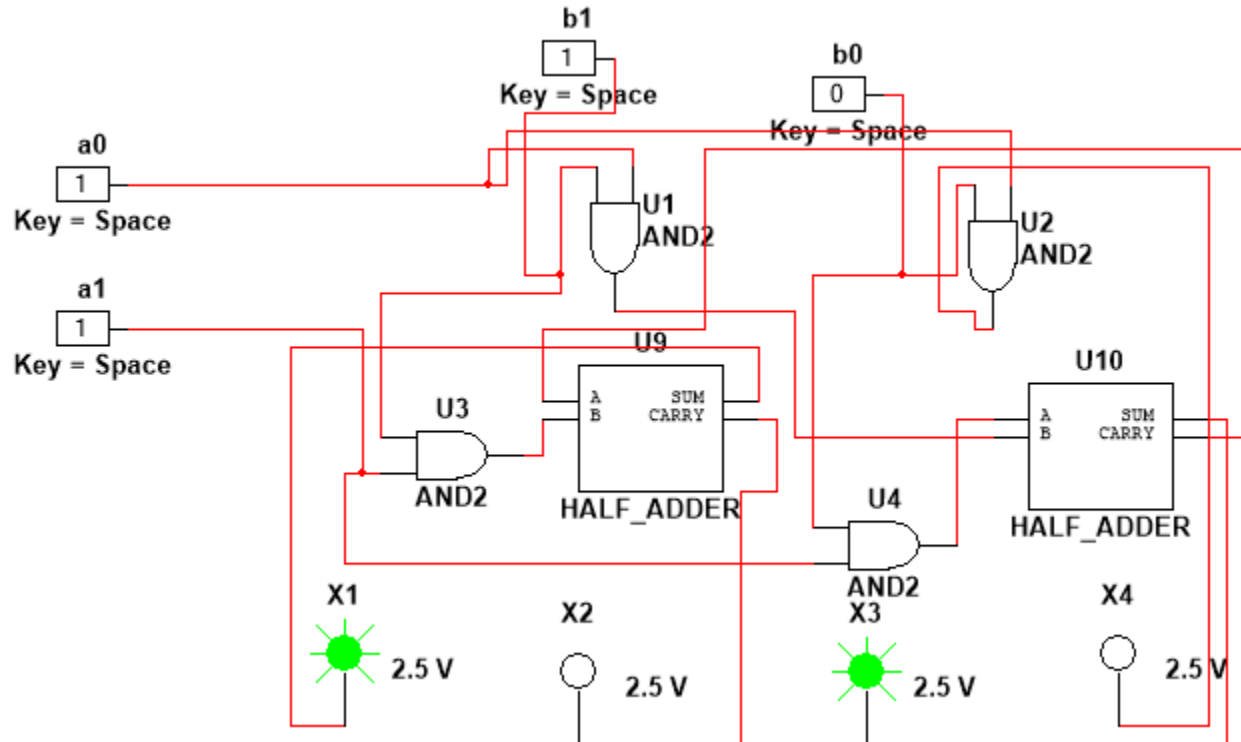
Circuit Diagram:



Truth table:

A1	A0	B1	B0	P3	P2	P1	P0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0

1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	1	0	0	1	1	0
1	1	0	0	0	0	0	0

Multisim Circuit:**Components required:**

1. Half adders
2. Interactive digital constants
3. probes

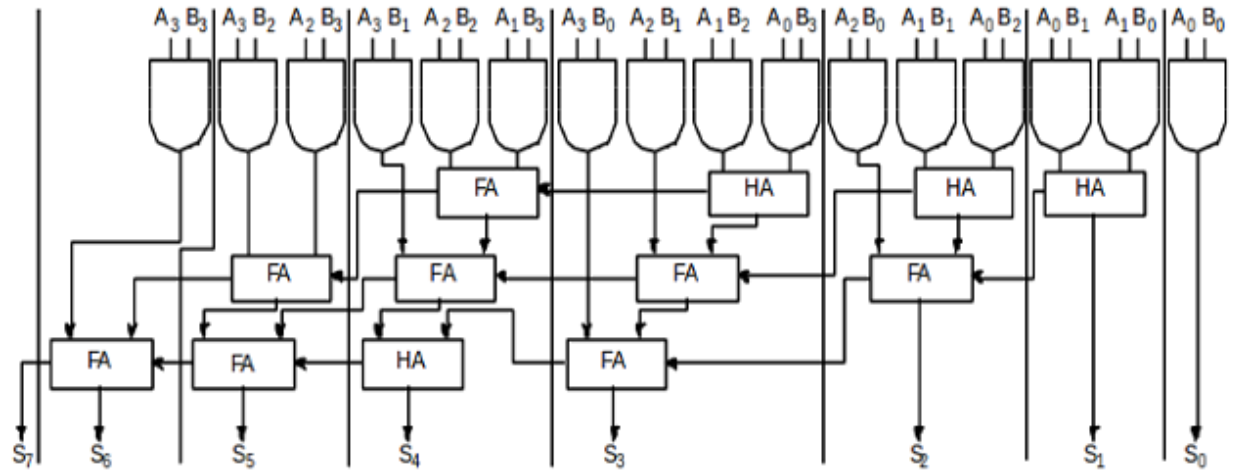
Procedure:

1. open multisim and create a new blank template.
2. Construct the circuit by connecting the input constants and the led (to the output).
3. verify the circuit diagram with the help of truth table

LAB-05

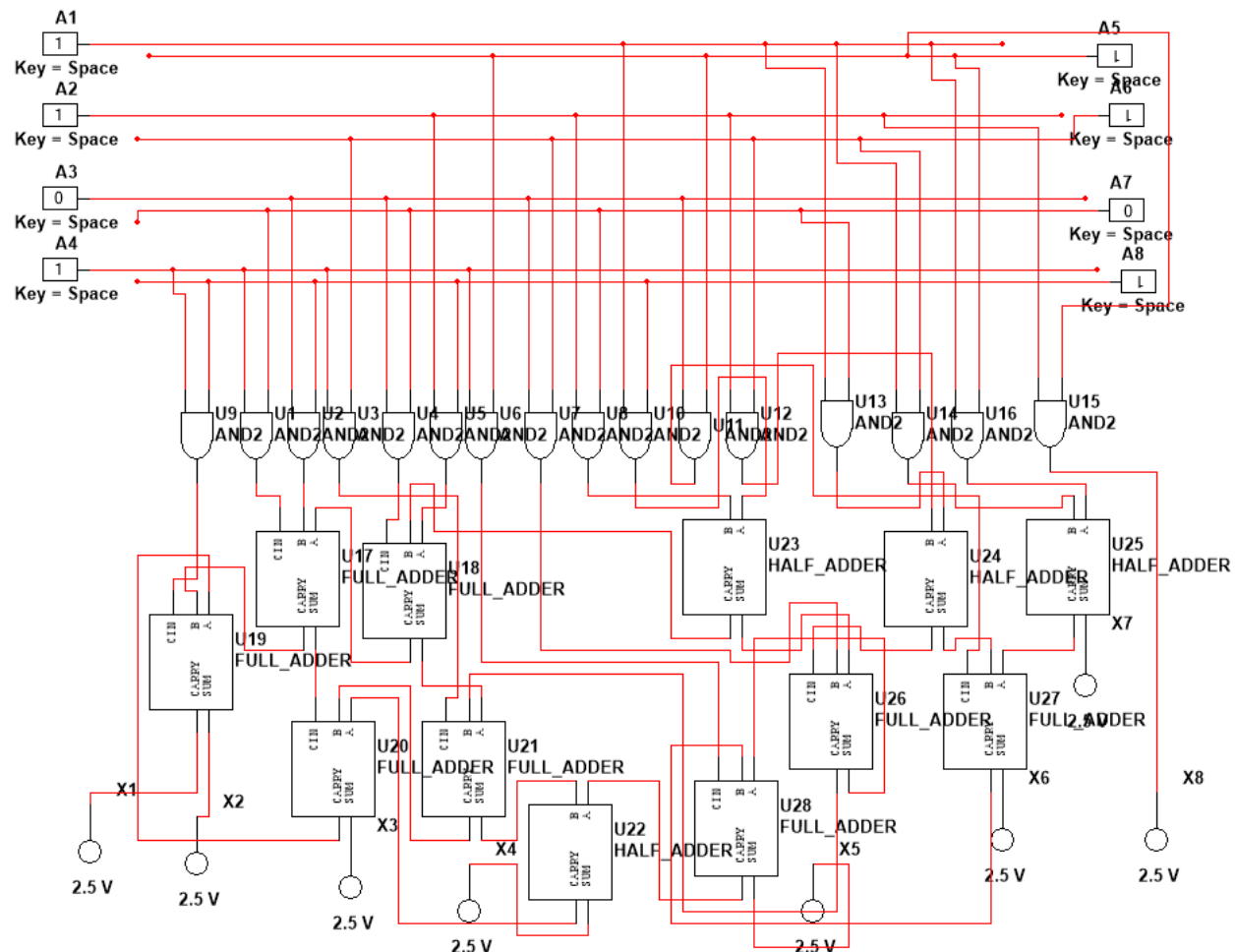
a) Aim: Verification of combinational Multiplier.

Description: A 4x4 combinational multiplier swiftly computes the product of two 4-bit binary numbers using logic gates, eliminating the need for sequential elements.

Circuit diagram:**Truth table:**

A				B				SUM							
A3	A2	A1	A0	B3	B2	B1	B0	S7	S6	S5	S4	S3	S2	S1	S0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1
0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	1	1	0	0	1	1	0	0	0	0	1	0	0	1
0	1	0	0	0	1	0	0	0	0	0	1	0	0	0	0
0	1	0	1	0	1	0	1	0	0	0	1	1	0	0	1
0	1	1	0	0	1	1	0	0	0	1	0	0	1	0	0
0	1	1	1	0	1	1	1	0	0	1	0	1	0	0	1
1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
1	0	0	1	1	0	0	1	0	1	0	1	0	0	0	1
1	0	1	0	1	0	1	0	0	1	1	0	0	1	0	0
1	0	1	1	1	0	1	1	0	1	1	1	1	0	0	1
1	1	0	0	1	1	0	0	1	0	0	1	0	0	0	0
1	1	0	1	1	1	0	1	1	0	1	0	1	0	0	1

1	1	1	0	1	1	1	0	1	0	1	0	0	0	0	0
1	1	1	1	1	1	1	1	1	0	1	0	1	0	0	1

Multisim Circuit:**Components required:**

1. 8 bit full adder
2. 4 half adder
3. 16 AND gates
4. 8 bit switch
5. 8 bit displays
6. Wires

Procedure:

1. To construct a 4x4 multiplier we need the components mentioned above.
2. Open the multisim and create a new schematic.

- 3.place the components and connect the circuit as shown in the circuit diagram.
- 4.run the circuit to see the result.
- 5.output is verified based on the truth table.